Profiting Off Prediction Markets – Using Machine Learning Models to Generate Profit off Insider Signals on Polymarket

Polymarket

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Introduction & Aim

Prediction markets, such as Polymarket, are peer-to-peer trading platforms that allow anyone to place bets on real-world events. As these markets have increased in popularity, research has found that:

1) The majority of winnings go to a small minority of wallets
2) Betters are generally more likely to bet "yes" than "no" outcomes
3) There have been numerous reports of insider trading.

This raises the question: are there psychological biases of betters or insider trading signals we can exploit in these markets? Our goal was to use models of varying complexity to detect these insider signals, and piggyback on these informed traders to generate profit.

Data Pipeline

All data was collected from the Polymarket API endpoints using a custom python script.

Selected market tags that have high potential for insiders: "Earnings", "Politics", "Acquisitions", "IPOs", and "Political Appointments".

Markets File
List of all closed markets for a given tag

History of trades executed in each market. Normalized to be Buy/Sell of the "Yes" side to maximize similarity to standard financial markets.

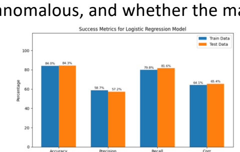
Trades File
Complete history of all trades in the market

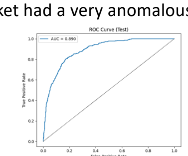
Relevant information about the wallets that are participating in each market:
"Total Trades", "Markets of Participation", "First Seen"

Wallet File
Metadata about market participants

Logistic Regression

The logistic regression model was used as our simplest baseline model to determine whether we can predict market outcomes with correlated insider trading features. Our model was trained on data from market open to 6 hours before close to determine if there was sufficient predictive signal before resolution. The features chosen were average trade size, difference in price from open to 6 hours before close, last trade price, percent of all trades that were anomalous, and whether the market had a very anomalous trade.

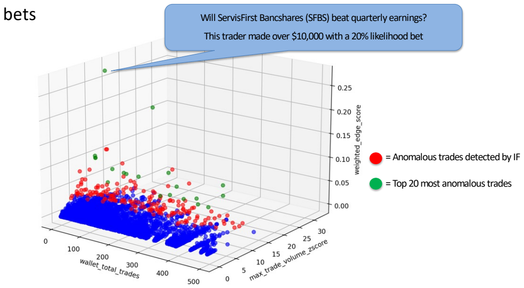




The model was (83%) accurate on the test set, which is 5 percentage points above a random guess baseline. The AUC-ROC score was 0.89, and the model found Last Price (-0.59) and Average Trade Size (3.19) to be the most important coefficients.

Isolation Forest

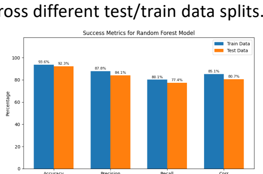
Overview: Isolation forest identifying anomalous trades using wallet's total trade count, Z-score of the max-volume trade in the market, and a weighted edge score representing profit from unlikely bets

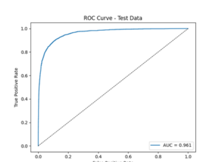


The aim of the Isolation Forest was to detect suspicious actors matching our idea of an insider trader. The model successfully isolated traders who extracted high profits from positions that were contrarian to the market consensus. This confirmed that suspicious behavior is detectable at the wallet level. However, at the individual trade level, the model did not show any clear indicators

Random Forest

The goal of the Random Forest was to provide a more advanced classification algorithm than the logistic regression model, using the same features to provide a more robust prediction of market outcomes (yes/no). Random Forests use an ensemble of decision trees (CARTs) to reduce variance and overfitting while capturing non-linear relationships between features. Adding K-fold cross validation further increases the reliability of the results by averaging across different test/train data splits.





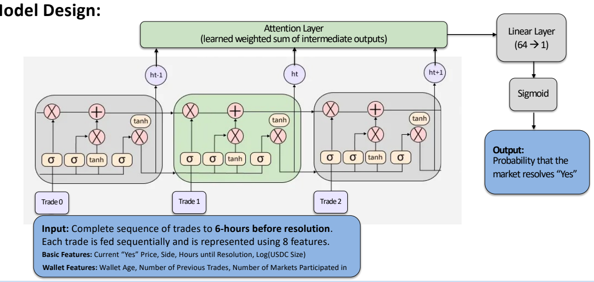
This model's accuracy (**92.3%**) significantly outperforms both the baseline (78%) and the logistic regression model (83%) when assessed using previously unseen markets (test data).

Model Hyperparameters:
Leaf size: 20, Ensemble size: 50, K: 5, Train/test split: 80/20%

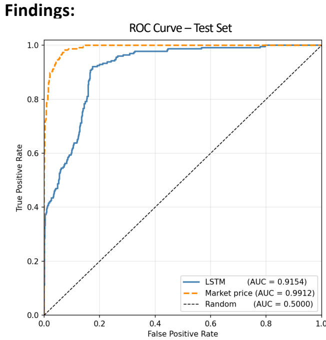
LSTM Model

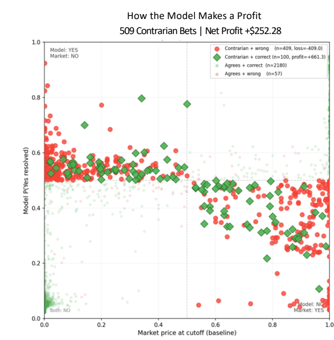
Overview:
The most complex model we trained was a Long Short-Term Memory (LSTM) Model. This variation of Recurrent Neural Network was selected because of its ability to incorporate both long-term data with short-term signals in a sequential and time-aware manner. A 70/30 train-test split was used (6749 train, 2746 test).

Model Design:



Findings:





The LSTM generated an **average ROI of 6.2%** across the 2746 test markets**.** This edge was generated not by simply relying on the feature with the highest correlation (current price with 6 hours to go) but through developing learned connections between the 8 input features and the LSTM cell outputs.

Conclusions

The models we implemented were able to profitably trade on features associated with insiders. Our isolation forest was able to identify highly suspicious users, and the LSTM was able to generate positive ROI by taking contrarian positions with high profit potential. Multiple models found that the closing price is highly coorelated with market outcomes, but additional experiments showed that a simple market price strategy is unprofitable. With 70% of profits going to only **0.4**% of wallets, Polymarket is an exploitable market, and the results of this project suggest basic machine learning models may be able to successfully generate profit.

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